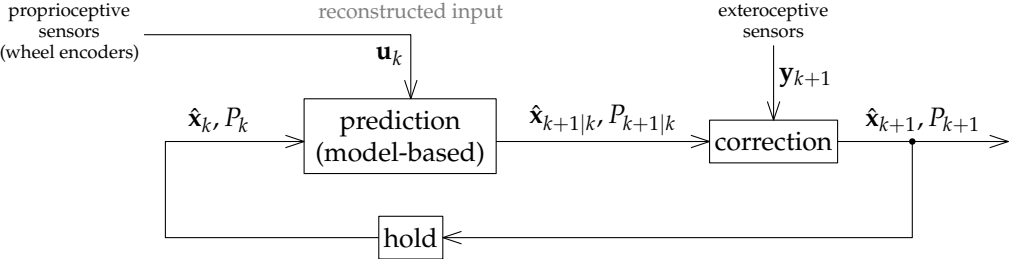
proprioceptive sensors (wheel encoders)
reconstructed input
exteroceptive sensors
$\mathbf{u}_k$
$\mathbf{y}_{k+1}$
$\hat{\mathbf{x}}_k, P_k$
prediction (model-based)
$\hat{\mathbf{x}}_{k+1|k}, P_{k+1|k}$
correction
$\hat{\mathbf{x}}_{k+1}, P_{k+1}$
hold